

Rossini Martyr

Data Science | Analytics

647-938-9055 | rossinimartyr@gmail.com | linkedin.com/in/rossinimartyr/ | github.com/Rossini-Martyr

MOTIVATION

I am a professional engineer, driven to **solve business problems** using Data Science & Analytics. I bring a systematic and collaborative approach to **add tangible value** to the team, business, and clients. I am a **life-long learner**, constantly seeking new ways to improve.

SKILLS & TOOLS

Programming: SQL, Python (Pandas, Numpy, Matplotlib, Scikit-Learn, Langchain)

Tools: Excel, Tableau, Github, Streamlit

Math: Linear Algebra, Statistics (Hypothesis Testing, AB Testing, Causal Inference)

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Gradient Boosted Trees, Retrieval Augmented Generation

PROJECTS

Florida Flood Damage Prediction ([Github](#))

- In order to quantify financial exposure due to flooding, I built a **Random Forest regression model** to predict the flood damage as a function of property cost using publicly available insurance claim data. The model predicted the damage ratio to **+/- 15%**, and could be utilized in the insurance industry, for example, in order to set premiums. I deployed the model using Streamlit to illustrate how this model could be applied in practice.

Ontario Curriculum Q&A Platform ([Github](#))

- I created a **Retrieval Augmented Generation platform** which answers questions on the Ontario elementary school curriculum using Langchain to automate ingestion of hundreds of pages of the curriculum. This technique could be applied to business use cases where there is a treasure trove of proprietary information which needs to be queried in order to collate information for downstream use.

Ontario EQAO Prediction Project ([Github](#))

- Ontario elementary schools are reporting record low standardized testing results in Mathematics. I built a **Random Forest Classifier** which identifies Ontario schools at risk of having a pass rate below 50% in the following year's EQAO exam. The model achieved a **Sensitivity of 74%**, which highlights the capability of the model to identify struggling schools. This can be used by the Ministry of Education to focus their limited resources on the schools which need it the most.

Ontario EQAO Segmentation Analysis ([Github](#))

- Used **K-Means Clustering** to segment over 3200 Ontario schools based on their EQAO performance in Mathematics. This information could be analyzed to identify the common features that the identified school segments have in common, thereby helping to inform the Ministry's education policy.

WORK EXPERIENCE

Senior Engineer, Fast + Epp (Toronto, Canada)

February 2024 – Present

- Project lead for the technical delivery of the \$120M New Brunswick Museum project, optimizing structural solutions to save over \$200,000 in material and resource efficiency
- Managed cross-functional teams on \$66M infrastructure projects, meeting stakeholder requirements while ensuring on-time and on-budget delivery of key deliverables
- Mentored junior and intermediate engineers, providing career guidance and subject matter review

Senior Engineer, Arup Canada (Toronto, Canada)

April 2016 – February 2024

- Developed an SQL and Python-based analytical workflow to process and analyze datasets with millions of rows, leveraging Amazon Redshift database and cloud compute power to accelerate and iterate design solutions
- Automated the creation of analytical models of underground stations in Toronto using Python-based API's, improving efficiency and accuracy by approximately 30%
- Supervised multiple junior engineers, conducting annual reviews, reviewing peer feedback and providing recommendations on career direction and promotion readiness

EDUCATION

MASc (Structural Engineering) (GPA: 3.94/4.00)

2013 - 2016

University of Toronto

- Published M.A.Sc research in prestigious structural engineering journal

BASc (Civil Engineering) (GPA: 3.88/4.00)

2008 - 2013

University of Toronto

- Graduated #1 in Civil Engineering

COURSES, CERTIFICATES AND QUALIFICATIONS

Data Science Professional Certification

2025 - 2026

IBM

- **Actionable Learnings:** Extracting & manipulating data using SQL. Automated web scraping using BeautifulSoup and web APIs. Utilising Github for version control, and collaboration. Using Python for data analysis, manipulation & visualization. Applying data preparation steps for ML including missing values, categorical variable encoding, outliers, feature scaling, feature selection & model validation. Applying Machine Learning algorithms for regression, classification, dimensionality reduction and clustering. Machine Learning pipelines to streamline the ML pre-processing & modelling phase. Deployment of an ML pipeline onto a live website using Plotly to create a Dash app. Using Plotly, Folium, and PyWaffle to create powerful Data Visualizations. Using Generative AI tools to assist in streamlining code production. Turning business problems into Data Science solutions.

Professional Engineer in Ontario

2018 - Current

PEO

- **Actionable Learnings:** As a licensed professional engineer in the province of Ontario, my foremost responsibility is to apply rigorous engineering methodology to ensure public safety in buildings and infrastructure projects